



UNIVERSITI TEKNOLOGI MALAYSIA

TASK 5

NETWORK COMMUNICATION

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Meeting Minutes #1

5.1 Get the network address from your lecture

Group	Network IP address
1	10.16.0.0/12
2	10.32.0.0/12
3	10.48.0.0/12
4	10.64.0.0/12
5	10.80.0.0/12
6	10.96.0.0/12
7	10.112.0.0/12
8	10.128.0.0/12
9	10.144.0.0/12
10	10.176.0.0/12

Figure 5.1.1

Figure 5.1.1 shows the list of network addresses of each group. The network address of our group is 10.112.0.0/12 . We assume that our subnet mask is /20 or 255.255.240.0.

Network Address details:

IP address (decimal)	10.112.0.0
IP address (binary)	00001010.01110000.00000000.00000000
Subnet mask (decimal)	255.255.240.0
Subnet mask (binary)	11111111.11111111.11110000.00000000

Network address	10.112.0.0
Broadcast address	10.112.15.255

Range of usable address	10.112.0.1 - : 10.112.15.254
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5.2.1 Network distribution for different work area

Utilizing the classful addressing approach, we opted to allocate distinct subnets based on the defined work areas in task 4. Consequently, our building's network necessitates the creation of 5 subnets. To achieve this, an additional 3 bits ($2^3 = 8$) must be borrowed from the host portion for subnet construction.

Subnet	Network portion	Subnet bits	Host portion		IP address
0	11000000 00010010	000	00000 00000000	Network Address	10.112.0.0
			11111 11111111	Broadcast Address	10.112.15.255
1	11000000 00010010	001	00000 00000000	Network Address	10.112.16.0
			11111 11111111	Broadcast Address	10.112.31.255
2	11000000 00010010	010	00000 00000000	Network Address	10.112.32.0
			11111 11111111	Broadcast Address	10.112.47.255
3	11000000 00010010	011	00000 00000000	Network Address	10.112.48.0
			11111 11111111	Broadcast Address	10.112.63.255
4	11000000 00010010	100	00000 00000000	Network Address	10.112.64.0
			11111 11111111	Broadcast Address	10.112.79.255

Distribution of subnet for different work area:

Subnet	Work Area	Network Address	Broadcast Address	Range of usable address	Subnet Mask Address
First Floor					
1	General Purpose Lab 1	10.112.0.0	10.112.15.255	10.112.0.1 - 10.112.15.254	/20 (255.255.240.0)
2	Hybrid Classroom	10.112.16.0	10.112.31.255	10.112.16.1 - 10.112.31.254	/20 (255.255.240.0)
3	Student Lounge	10.112.32.0	10.112.47.255	10.112.32.1 - 10.112.47.254	/20 (255.255.240.0)
4	Video Conferencing room	10.112.48.0	10.112.63.255	10.112.48.1 - 10.112.63.254	/20 (255.255.240.0)
Second Floor					
5	General Purpose Lab 2	10.112.64.0	10.112.79.255	10.112.64.1 - 10.112.79.254	/20 (255.255.240.0)
6	Embedded Lab	10.112.80.0	10.112.95.255	10.112.80.1 - 10.112.95.254	/20 (255.255.240.0)
7	Cisco Network Lab	10.112.96.0	10.112.111.255	10.112.96.1 - 10.112.111.254	/20 (255.255.240.0)
8	Cafe	10.112.112.0	10.112.127.255	10.112.112.1 - 10.112.127.254	/20 (255.255.240.0)

5.2.2 Network distribution for workstation in different work areas

First Floor:

General Purpose Lab 1

Devices	IP Address
PC 1	10.112.0.1
PC 2	10.112.0.2
PC 3	10.112.0.3
PC 4	10.112.0.4
PC 5	10.112.0.5
PC 6	10.112.0.6
PC 7	10.112.0.7
PC 8	10.112.0.8
PC 9	10.112.0.9
PC 10	10.112.0.10
PC 11	10.112.0.11
PC 12	10.112.0.12
PC 13	10.112.0.13
PC 14	10.112.0.14
PC 15	10.112.0.15
PC 16	10.112.0.16
PC 17	10.112.0.17
PC 18	10.112.0.18
PC 19	10.112.0.19
PC 20	10.112.0.20
PC 21	10.112.0.21
PC22	10.112.0.22

PC 23	10.112.0.23
PC 24	10.112.0.24
PC 25	10.112.0.25
PC 26	10.112.0.26
PC 27	10.112.0.27
PC 28	10.112.0.28
PC 29	10.112.0.29
PC 30	10.112.0.30

Hybrid Classroom

Devices	IP Address
PC 1	10.112.16.1
PC 2	10.112.16.2
PC 3	10.112.16.3
PC 4	10.112.16.4
PC 5	10.112.16.5
PC 6	10.112.16.6
PC 7	10.112.16.7
PC 8	10.112.16.8
PC 9	10.112.16.9
PC 10	10.112.16.10
PC 11	10.112.16.11
PC 12	10.112.16.12
PC 13	10.112.16.13
PC 14	10.112.16.14
PC 15	10.112.16.15

PC 16	10.112.16.16
PC 17	10.112.16.17
PC 18	10.112.16.18
PC 19	10.112.16.19
PC 20	10.112.16.20
PC 21	10.112.16.21
PC22	10.112.16.22
PC 23	10.112.16.23
PC 24	10.112.16.24
PC 25	10.112.16.25
PC 26	10.112.16.26
PC 27	10.112.16.27
PC 28	10.112.16.28
PC 29	10.112.16.29
PC 30	10.112.16.30

Video Conferencing Room

Devices	IP Address
PC 1	10.112.48.1

Second Floor:

General Purpose Lab 2

Devices	IP Address
PC 1	10.112.64.1
PC 2	10.112.64.2
PC 3	10.112.64.3

PC 4	10.112.64.4
PC 5	10.112.64.5
PC 6	10.112.64.6
PC 7	10.112.64.7
PC 8	10.112.64.8
PC 9	10.112.64.9
PC 10	10.112.64.10
PC 11	10.112.64.11
PC 12	10.112.64.12
PC 13	10.112.64.13
PC 14	10.112.64.14
PC 15	10.112.64.15
PC 16	10.112.64.16
PC 17	10.112.64.17
PC 18	10.112.64.18
PC 19	10.112.64.19
PC 20	10.112.64.20
PC 21	10.112.64.21
PC22	10.112.64.22
PC 23	10.112.64.23
PC 24	10.112.64.24
PC 25	10.112.64.25
PC 26	10.112.64.26
PC 27	10.112.64.27
PC 28	10.112.64.28

PC 29	10.112.64.29
PC 30	10.112.64.30

Embedded Lab

Devices	IP Address
PC 1	10.112.80.1
PC 2	10.112.80.2
PC 3	10.112.80.3
PC 4	10.112.80.4
PC 5	10.112.80.5
PC 6	10.112.80.6
PC 7	10.112.80.7
PC 8	10.112.80.8
PC 9	10.112.80.9
PC 10	10.112.80.10
PC 11	10.112.80.11
PC 12	10.112.80.12
PC 13	10.112.80.13
PC 14	10.112.80.14
PC 15	10.112.80.15
PC 16	10.112.80.16
PC 17	10.112.80.17
PC 18	10.112.80.18
PC 19	10.112.80.19
PC 20	10.112.80.20
PC 21	10.112.80.21

PC22	10.112.80.22
PC 23	10.112.80.23
PC 24	10.112.80.24
PC 25	10.112.80.25
PC 26	10.112.80.26
PC 27	10.112.80.27
PC 28	10.112.80.28
PC 29	10.112.80.29
PC 30	10.112.80.30
PC 31	10.112.80.31
PC 32	10.112.80.32
PC 33	10.112.80.33

Cisco Network Lab

Devices	IP Address
PC 1	10.112.96.1
PC 2	10.112.96.2
PC 3	10.112.96.3
PC 4	10.112.96.4
PC 5	10.112.96.5
PC 6	10.112.96.6
PC 7	10.112.96.7
PC 8	10.112.96.8
PC 9	10.112.96.9
PC 10	10.112.96.10
PC 11	10.112.96.11

PC 12	10.112.96.12
PC 13	10.112.96.13
PC 14	10.112.96.14
PC 15	10.112.96.15
PC 16	10.112.96.16
PC 17	10.112.96.17
PC 18	10.112.96.18
PC 19	10.112.96.19
PC 20	10.112.96.20
PC 21	10.112.96.21
PC22	10.112.96.22
PC 23	10.112.96.23
PC 24	10.112.96.24
PC 25	10.112.96.25
PC 26	10.112.96.26
PC 27	10.112.96.27
PC 28	10.112.96.28
PC 29	10.112.96.29
PC 30	10.112.96.30
PC 31	10.112.96.31
PC 32	10.112.96.32
PC 33	10.112.96.33

MEETING MINUTE 1

DATE/TIME		24th Disember 2024	
LOCATION		ONLINE - GMEET	
AGENDA		1) Discuss the question given for Task 5 2) Divide task for each member	
Meeting MC		YASMIN BATRISYIA BINTI ZAHIRUDDIN	
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
MUHAMMAD MUKHRITZ AL IMAN BIN MOHD RAFFI		2:00 PM	-
AHMAD ADIB ZIKRI BIN A.MAZLAM		2:00 PM	-
YASMIN BATRISYIA BINTI ZAHIRUDDIN		2:00 PM	-
SYARIFAH DANIA BINTI SYED ABU BAKAR		2:00 PM	-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discuss the materials given for Task 5	i) Syarifah Dania suggested all of the team members go through the questions and determine the requirements needed to complete Task5 . ii) All members read the materials given and identify the requirements	All members (25/12)

2	Network distribution	i) All of group member discuss the network distribution for our room ii) Giving task between member depend on their schedule to ensure task will be completed on time.	Yasmin Batrisyia (25/12) Muhammad Mukhritz Al Iman (25/12)
3	Meeting ended	After completing the dividing task between each member, Ahmad Adib ended the meeting.	Ahmad Adib (25/12)